

Short communication

Antimicrobial susceptibility and genetic features of a heterogeneous vancomycin intermediate-resistant *Staphylococcus aureus* strain

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ARTICLE INFO

Keywords:

Heterogeneous vancomycin-intermediated
Staphylococcus aureus
Genetic characteristics
Antimicrobial resistance
Virulence factors

ABSTRACT

This study aimed to characterize the antimicrobial susceptibility and genetic features of a heterogeneous vancomycin-intermediate *Staphylococcus aureus* (hVISA) strain Guangzhou-SauVS2 recovered from a female patient in Guangzhou, representative of southern China. The genome of Guangzhou-SauVS2 was sequenced using Illumina HiSeq 2500 platform and assembled de novo using Velvet v1.2.08. Annotations and bioinformatics analysis were further performed. Results showed that Guangzhou-SauVS2 was susceptible and resistant to 7 and 11 antibiotic drugs, respectively, and exhibited hVISA with a minimum inhibitory concentration of vancomycin as 4 µg/mL. Its genome is 2,883,941 bp in length and contains 2934 predicted genes with an average G + C content of 32.9%. Besides, a total of 38 virulence factors and 4 antibiotic-resistant genes were identified. These results can be employed to further study the pathogenic and antimicrobial mechanisms of hVISA

1. Introduction

Staphylococcus aureus is a well-known pathogen causing food spoilage and diverse infectious diseases due to its ability to adapt to different environments, especially for antibiotic resistance (Biliaderis et al., 1986; Control and Prevention, 2003; Gardner et al., 2018; Murray, 2005). Methicillin-resistant *S. aureus* (MRSA) has been spread worldwide since the 1980s and tended to increase in recent years (Chen et al., 2010). Treatment of serious MRSA infections often involves the glycopeptide antibiotic vancomycin. Unfortunately, heterogeneous vancomycin-intermediated *S. aureus* (hVISA), was firstly detected in Japan (Hiramatsu et al., 1997) and then clinical isolates had been reported around the world (Howden et al., 2010; Smith et al., 1999), which poses a serious challenge to the control and treatment of MRSA.

In China, the prevalence of hVISA among MRSA isolates from central-southern China in 2011–2012 was 22.1%, which was the highest in Asia (Liu et al., 2015). In the present study, we isolated a hVISA strain Guangzhou-SauVS2 from southern China, and its antimicrobial susceptibility and genetic features, including virulence and antibiotic-resistance genes, were investigated.

2. Materials and methods

The Guangzhou-SauVS2 was isolated from the urine of a 49-year-old woman who was diagnosed as urinary tract infection with none of vancomycin treatment history in the secretory ward of the First Affiliated Hospital of Guangzhou Medical University in Guangzhou (China) in 2014. Bacterial identification and MIC testing of 19

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antimicrobial agents were determined as previously described with VITEK 2™ Automated System and Etest Strips (Xu et al., 2018). For confirmation of hVISA, modified vancomycin population analysis profile/area under the curve (PAP/AUC) was performed as previously described (Wootton et al., 2001) using Mu3 (ATCC 700698) and MU50 (ATCC 700699) as reference strains. The draft genome of Guangzhou-SauVS2 was sequenced using the Illumina HiSeq 2,500 platform and paired-end libraries. The quality control of reads was carried out (Liu et al., 2016). Afterwards, clean data gained were assembled de novo using Velvet v1.2.08 (Zerbino and Birney, 2008). Gene predictions were performed using Glimmer 3.0 (Delcher et al., 2007). Predicted genes were further annotated by BLAST (<https://blast.ncbi.nlm.nih.gov/Blast.cgi>), GO database (<http://www.geneontology.org/>) (Ashburner et al., 2000), KOG database (<https://www.ncbi.nlm.nih.gov/COG/>) (Tatusov et al., 2003), and KEGG pathway database (<http://www.genome.jp/kegg/>) (Kanehisa and Goto, 2000). Transfer RNA (tRNA) was predicted by tRNA scan-SE (v1.21) (Avni et al., 1997). Importantly, virulence factors were predicted by using the core database of Virulence Factors of Pathogenic Bacteria Database (VFDB, <http://ardb.cccb.umd.edu/>), and antibiotic-resistance genes were predicted through Resfinder 3.0 (Zankari et al., 2012).

3. Results

The Guangzhou-SauVS2 showed resistance against β -lactams (including Penicillin, Oxacillin, Cefuroxime, Cefazolin, and Cefoxitin), quinolones (including Moxifloxacin, Ciprofloxacin, and Levofloxacin), macrolides (Erythromycin), tetracyclines (Tetracycline), and Lincosamides (Clindamycin). Besides, it exhibited VISA with a MIC of vancomycin as 4 μ g/mL (Table 1). PAP/AUC was then performed to confirm its antibiotic susceptibility pattern and it turned out that the AUC ratio of Guangzhou-SauVS2 to MU3 was 0.95 (Fig. 1), therefore, Guangzhou-SauVS2 was defined as hVISA.

The final assembly of Guangzhou-SauVS2 yielded 184 scaffolds with a total length of 2,883,941 bp and an average G + C content of 32.9%, in which the largest scaffold was 95,802 in length and N50 scaffold size was 16,031 bp. The coverage of the whole genome was 99%. Additionally, a total of 2934 predicted genes (including 3 tRNA genes and 2696 protein-coding genes) were identified, in which 952 predicted genes, 2419 predicted genes, and 1120 predicted genes were assigned

Table 1

Antimicrobial susceptibility identified in *Staphylococcus aureus* strain Guangzhou-SauVS2.

Antimicrobial agents	MIC (μ g/mL)	Susceptibility ^a
Penicillin	≥ 0.5	R
Oxacillin	≥ 4	R
Cefuroxime	6	R
Cefazolin	6	R
Cefoxitin	6	R
Gentamycin	≤ 0.5	S
Moxifloxacin	4	R
Ciprofloxacin	≥ 8	R
Levofloxacin	≥ 8	R
Erythromycin	≥ 8	R
Tetracycline	≥ 16	R
Tigecycline	0.5	S
Vancomycin	4	I
Quinupristin/dalfopristin	0.5	S
Clindamycin	≥ 8	R
Linezolid	2	S
Rifampicin	≤ 0.5	S
Nitrofurantoin	≤ 16	S
Trimethoprim/sulfamethoxazole	≤ 10	S

MIC, minimum inhibitory concentration; R, resistant; S, susceptible; I, intermediate.

^a Interpreted according to Clinical and Laboratory Standards Institution (CLSI) guidelines for *S. aureus*.

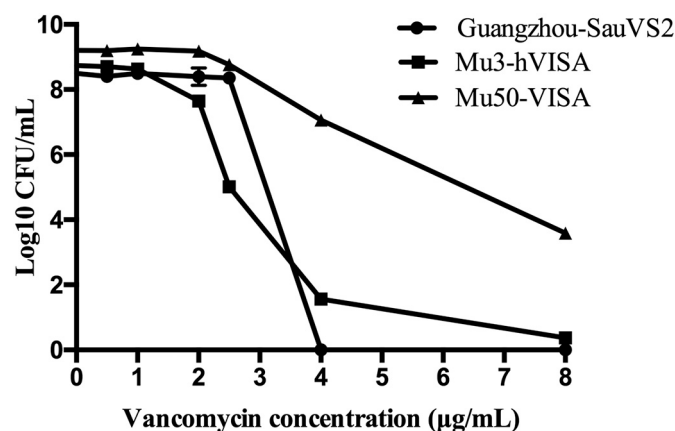


Fig. 1. Population analysis profile curves of Guangzhou-SauVS2 and two control strains (VISA strain Mu50 and hVISA strain MU3).

into GO function annotations (Fig. 2), KOG function annotations (Fig. 3), and KEGG pathway (Fig. 4), respectively. Furthermore, 38 virulence factors were identified in the genome of Guangzhou-SauVS2 (Table 2). The largest number stemmed from 13 *cap* genes (34.2%) encoding capsular polysaccharide synthesis enzyme. Followed by 10 genes (26.3%), including four *ess* genes, three *esa* genes, and three *esx* encoding type VII secretion system protein. Considering the potential antibiotic resistance, 4 different types of genes were identified, namely, *tet* (*M*) (11,914–13,833 bp), *mecA* (3406–5415 bp), *spc* (2316–3098 bp) and *erm* (*A*) (1459–2190 bp).

4. Discussion

The Guangzhou-SauVS2 was susceptible to two antistaphylococcal agents (linezolid and tigecycline), besides, it showed drug resistance to multiple antibiotics, including levofloxacin, erythromycin, and tetracycline (Table 1). These results correlated well with a former study on clinical isolates of hVISA in China (Liu et al., 2015).

Based on the KEGG pathway, the highest number of genes in the draft genome of Guangzhou-SauVS2 were involved in metabolism, within the metabolism, metabolic pathways comprised of the highest number of genes at 156 followed by biosynthesis of secondary metabolites at 77 genes, and biosynthesis of antibiotics at 67 genes. Moreover, there were 22 and 8 predicted genes involved in purine metabolism and pyrimidine metabolism, respectively. Former studies suggested that increased purine biosynthesis and pyrimidine metabolism could enhance ATP production and peptidoglycan synthesis, respectively, which both lead to increased cell wall thickness for VISA strains (Howden et al., 2011; Mongodin et al., 2003). It is worthy to note that two genes were encoding for vancomycin resistance, in which one is involved in D-alanine biosynthetic process, while the other is responsible for peptidoglycan biosynthetic process and cell wall organization. Among environmental information processing, ABC transporters consisted of the highest number of genes at 42 followed by two-component systems (TCSs) at 25 and phosphotransferase system at 15. ABC transporters and TCSs allow *S. aureus* to detect and make quick responses to the environmental signs, which contribute to its survival in tough environments (Falord et al., 2012).

The clinical outcome of hVISA is closely related to its virulence factors and antibiotic-resistant genes. The functional properties of virulence factors of *S. aureus* could be characterized as adherence, immune invasion, and invasion according to their functional properties (Carrera et al., 2017; Zecconi and Scali, 2013). In the genome of Guangzhou-SauVS2, 7 virulence factors were involved in adhesion to host cells; 14 virulence factors were involved in immune invasion, including *cap* and *adsA* genes. Capsular polysaccharide encoded by *cap* can interact with the host immune system and protect the organism

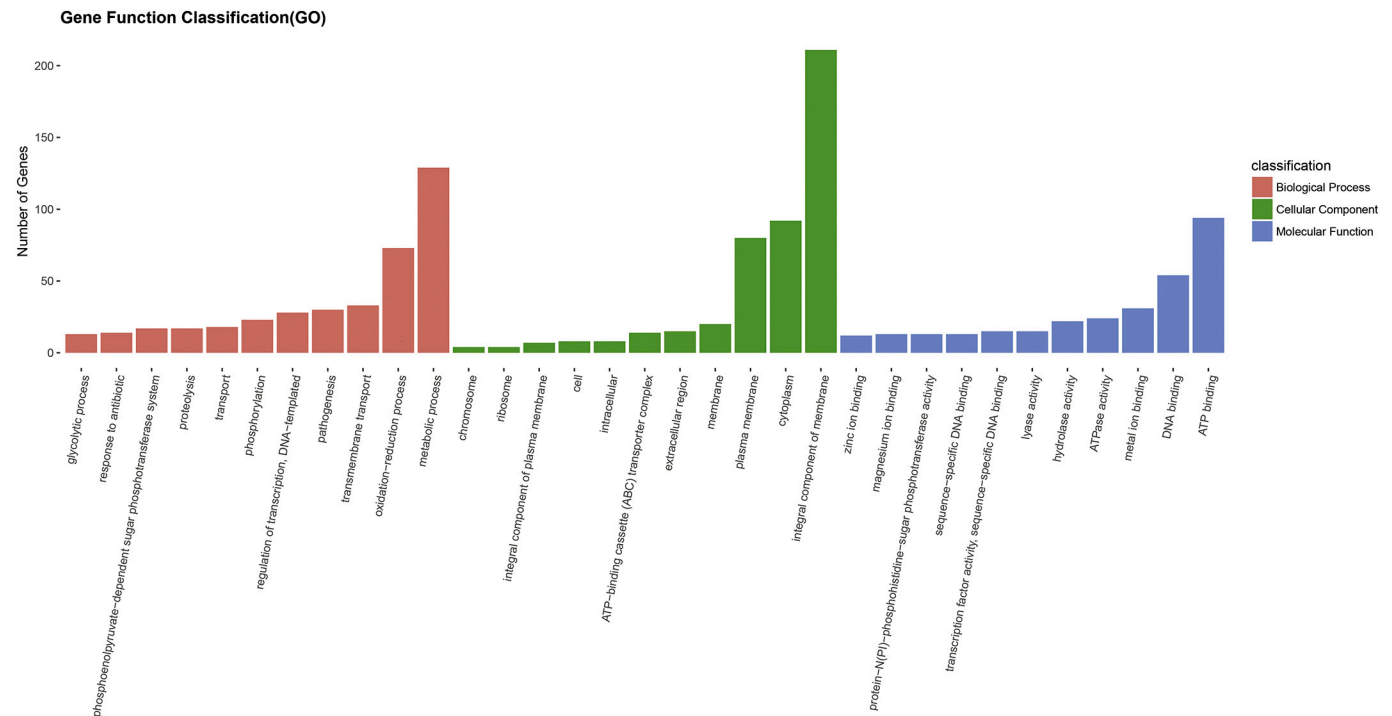


Fig. 2. GO terms enrichment of genes in Guangzhou-SauVS2.

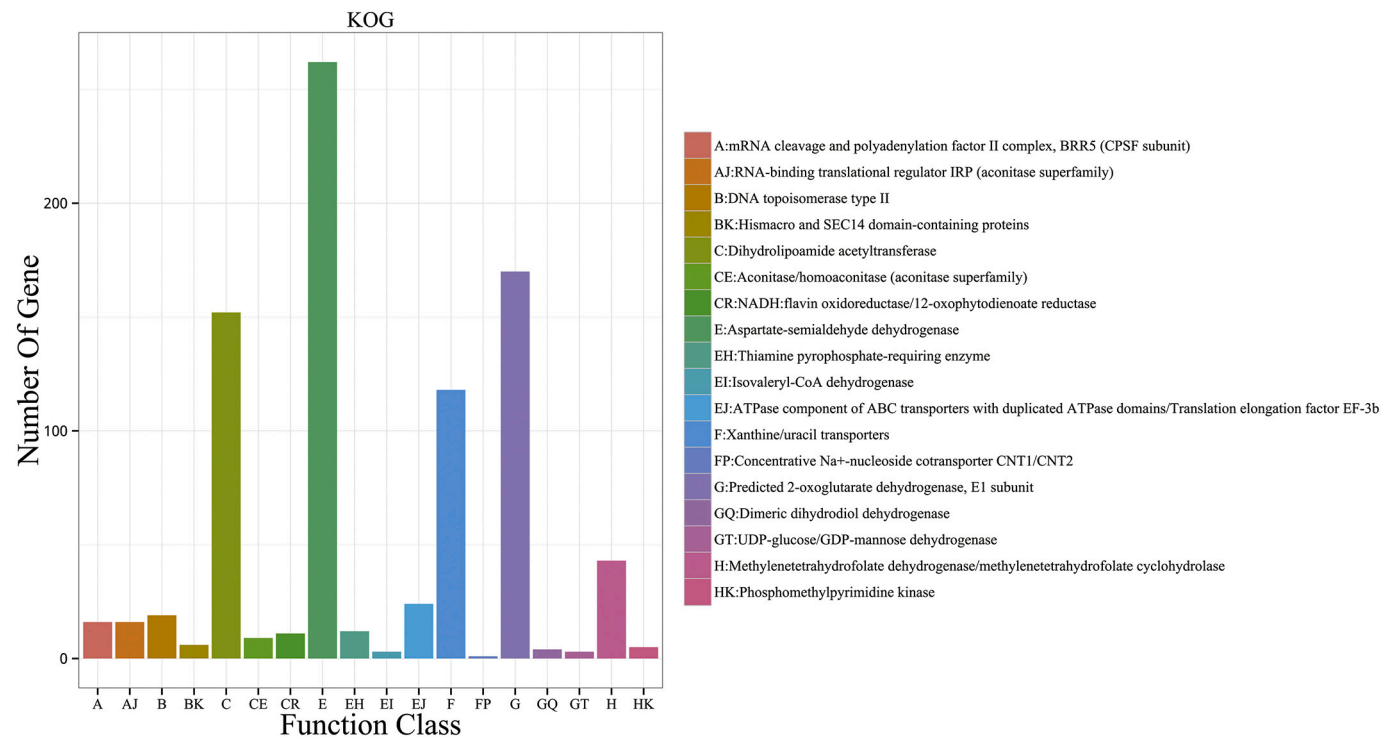


Fig. 3. KOG function classification in Guangzhou-SauVS2.

against being captured and killed by phagocytes (Zhao et al., 2010). A cell wall anchored protein encoded by *adsA* could be used by *S. aureus* to increase the concentration of adenosine to escape immune clearance due to its immunosuppressive properties (Thammavongsa et al., 2011). Besides, 12 virulence factors were identified as invasion virulence factors. However, it should be noted that the most frequent toxin genes detected in hVISA by polymerase chain reaction were hemolysin genes (except *hlg*) followed by *lukED* (Liu et al., 2015). Four identified genes

involved in antibiotic resistance correlated well with the antimicrobial susceptibility results shown in Table 1. *Tet* (*M*) gene confers resistance to tetracycline by interacting with ribosomes and enhancing the release of tetracycline (Burdett, 1991). *MecA* gene is responsible for beta-lactam resistance because it codes penicillin-binding protein 2A which presents a low affinity to all beta-lactam antibiotics (Utsui and Yokota, 1985). *Spc* gene makes the antibiotic fail to bind with ribosomes through encoding spectinomycin adenyl transferase (Shaw et al., 1993).

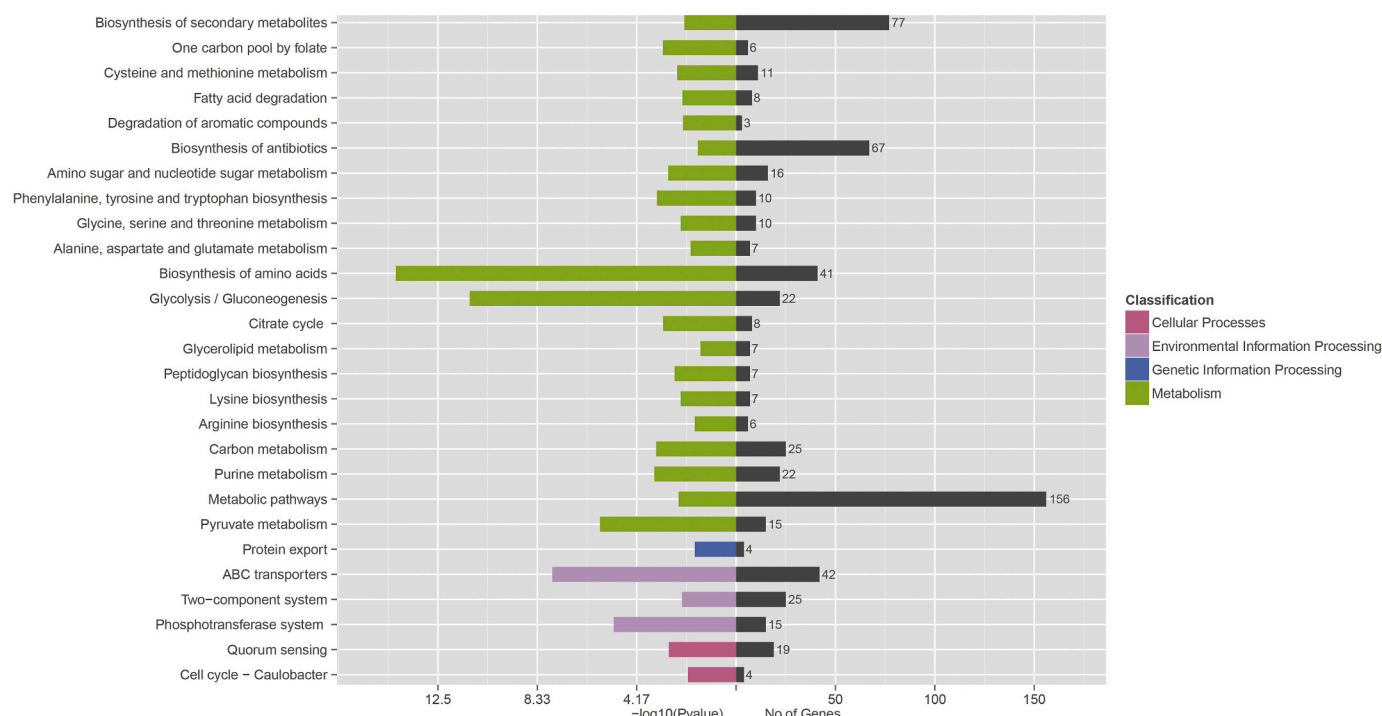


Fig. 4. KEGG pathway enrichment results of genes in Guangzhou-SauVS2.

Table 2
Conclusion of genes involved in virulence of Guangzhou-SauVS2.

Functional category	Description	Virulence genes	Identity (%)
Immune invasion	Capsular polysaccharide synthesis enzyme	<i>cap8A</i>	99.807
		<i>cap8B</i>	98.981
		<i>cap8C</i>	99.606
		<i>cap8D</i>	99.726
		<i>cap8D</i>	100
		<i>cap8E</i>	99.223
		<i>cap8F</i>	99.459
		<i>cap8G</i>	99.556
		<i>cap8L</i>	99.502
		<i>cap8M</i>	99.642
		<i>cap8N</i>	99.437
		<i>cap8O</i>	99.05
		<i>cap8P</i>	99.564
		<i>adsA</i>	96.421
Immune invasion Invasion	Adenosine synthase A Type VII secretion system protein	<i>essA</i>	98.257
		<i>essA</i>	100
		<i>essB</i>	100
		<i>essC</i>	98.911
		<i>esaA</i>	99.588
		<i>esaB</i>	100
		<i>esaB</i>	99.026
		<i>esxA</i>	98.514
		<i>esxB</i>	97.71
		<i>esxC</i>	97.46
Invasion	Staphopain cysteine proteinase	<i>sspB</i>	99.746
		<i>sspC</i>	99.697
		<i>clfA</i>	92.901
Adherence/ Immune invasion	Clumping factor A, fibrinogen-binding protein	<i>clfA</i>	89.488
		<i>clfA</i>	94.186
		<i>sdrC</i>	93.289
Adherence	Ser-Asp rich fibrinogen-binding bone sialoprotein-binding protein	<i>sdrD</i>	91.924
		<i>sdrE</i>	100
		<i>vWbp</i>	80.064
Adherence	Secreted von Willebrand factor-binding protein precursor	<i>sspA</i>	99.225
		<i>sspA</i>	98.75
		<i>coa</i>	92.418
Others	Staphylocoagulase precursor	<i>coa</i>	100
		<i>coa</i>	100
		<i>geh</i>	99.181
Others	Glycerol ester hydrolase	<i>geh</i>	99.181
		<i>geh</i>	99.181

Erm(A) gene confers resistance to macrolide through preventing the antimicrobial agents from reaching the drug-binding site on the ribosome (Maranan et al., 1997).

5. Conclusion

This study studied the antimicrobial susceptibility and genetic characteristics of a hVISA strain Guangzhou-SauVS2 through draft genome sequencing and bioinformatics analyses. The Guangzhou-SauVS2 was susceptible and resistant to 7 and 11 antimicrobial agents, respectively. Besides, its genome has a length of 2,883,941 bp and contains 2934 predicted genes with an average G + C content of 32.9%, with 38 virulence factors and 4 antibiotic-resistant genes identified. However, future studies involving the epidemiological properties and genetic traits related to the hVISA phenotype should be carried out to further investigate its pathogenic and antimicrobial mechanisms.

Funding

This work was supported by the National Key R&D Program of China (2016YFD0400203), YangFan Innovative and Entrepreneurial Research Team Project (2014YT02S029), the National Key Research and Development Program of China (2016YFD04012021), the Fundamental Research Funds for the Central Universities (Dr. Zhenbo Xu, D2191310), Collaborative grant with AEIC (KEO-2019-0624-001-1).

Declaration of Competing Interest

None to declare.

Acknowledgements

We sincerely thanks Dr. Hui Wang, a professor at the Peking University People's Hospital, for her donation of MU3 and MU50.

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